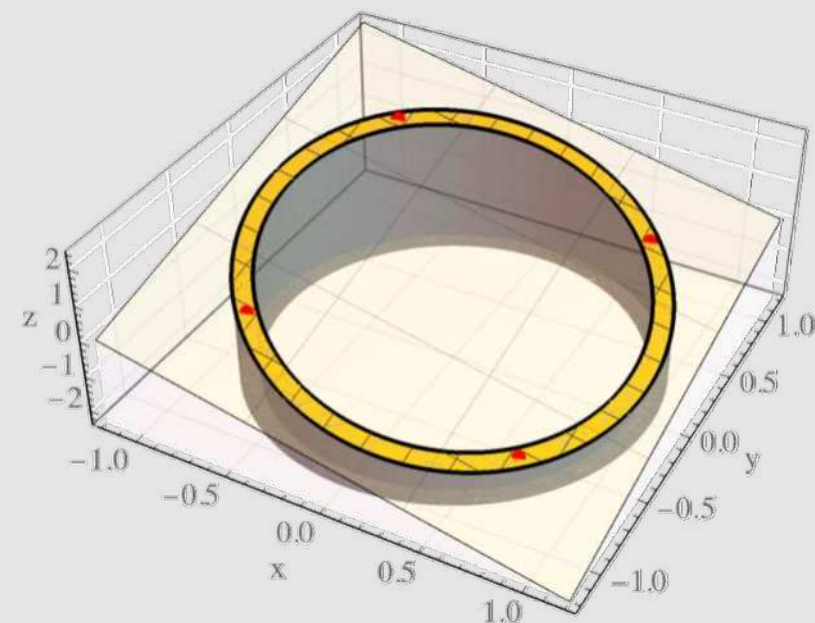
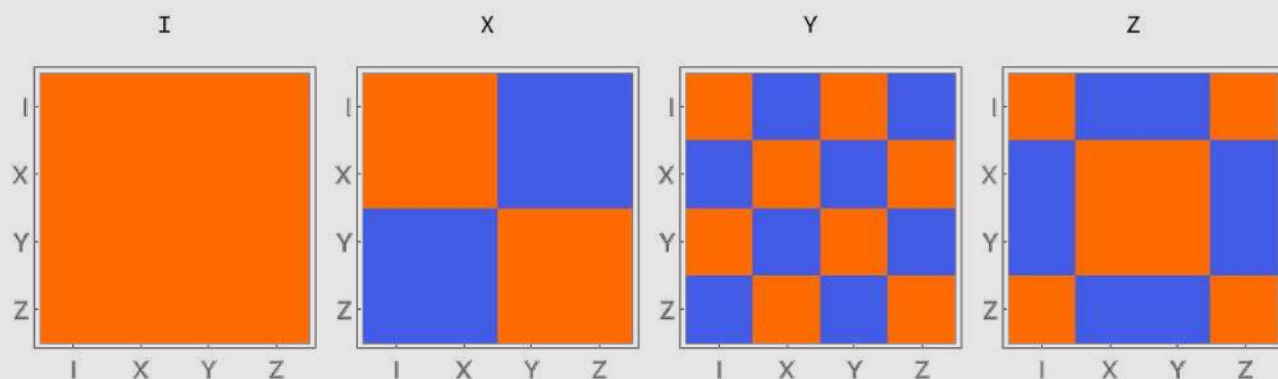
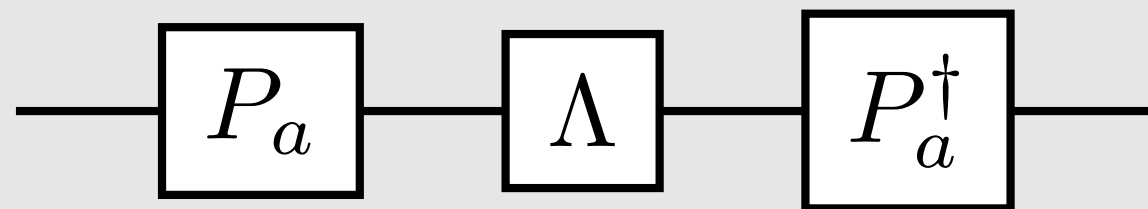


Primer on Pauli Twirling



Zlatko Minev

2022-04-20, 07-11

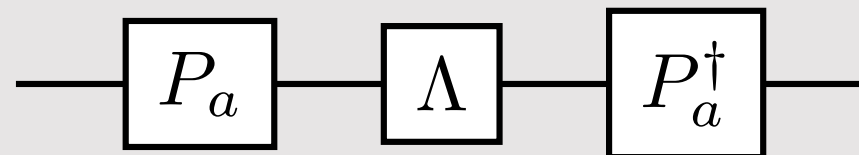
Twirling 101: Overview

Twirl operationally

Simple example

General application

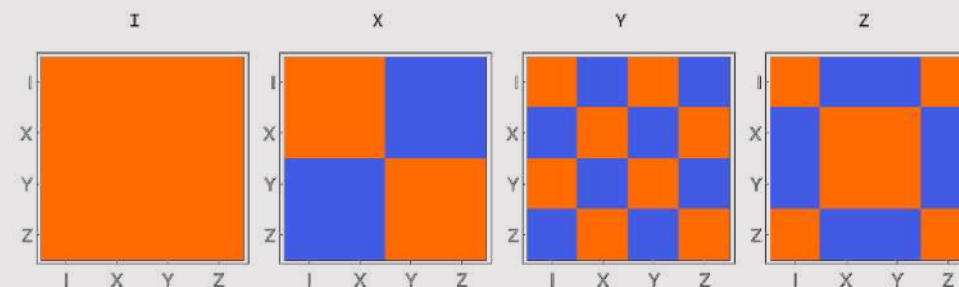
Summary



Theory of twirling

Why does twirling work?

Masking channels

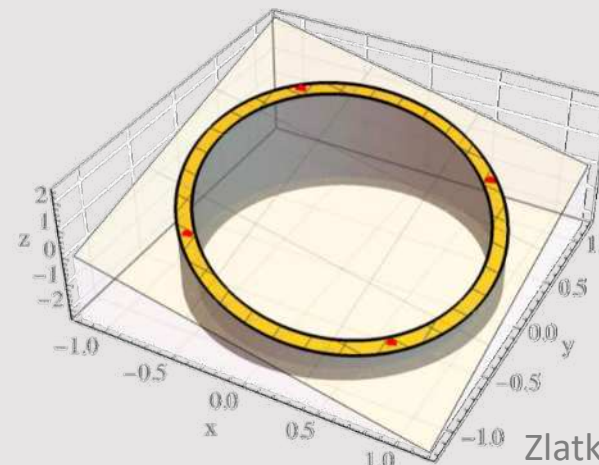


Optional: Advanced

Why is the Pauli group special for twirling?

Other twirl groups

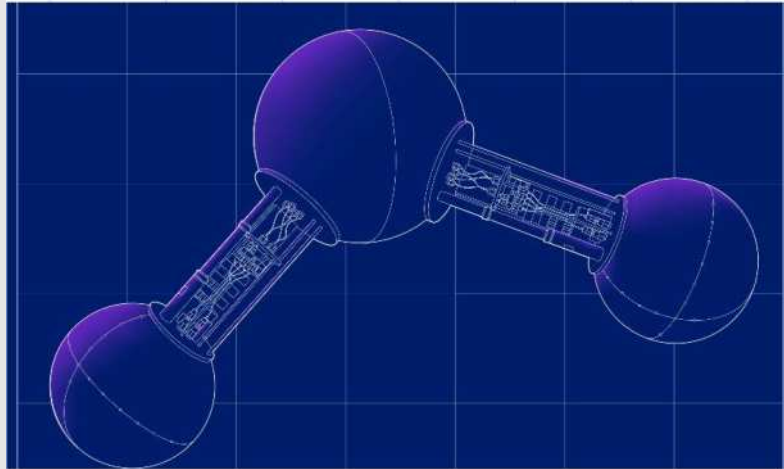
Designs



Why does twirling actually work?

Theory and my take on it

Noise basics 101



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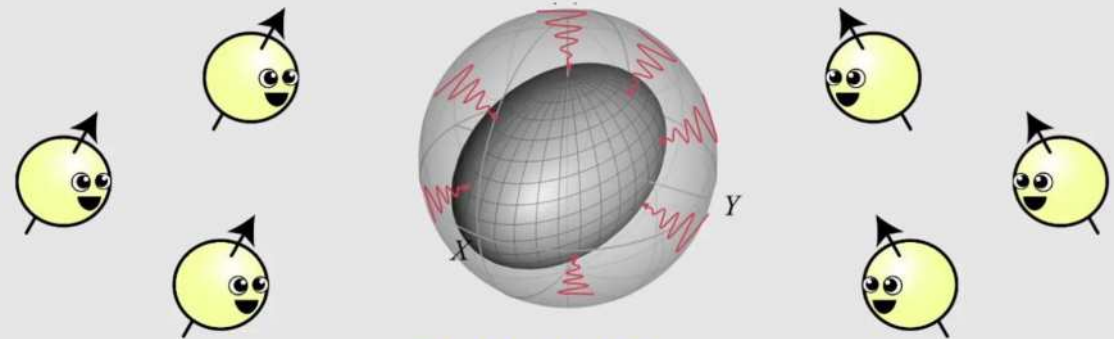
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Introduction to Quantum Noise

Qiskit Global Summer School: Quantum Simulations



Zlatko K. Minev
IBM Quantum

 @zlatko_minev  zlatko-minev.com



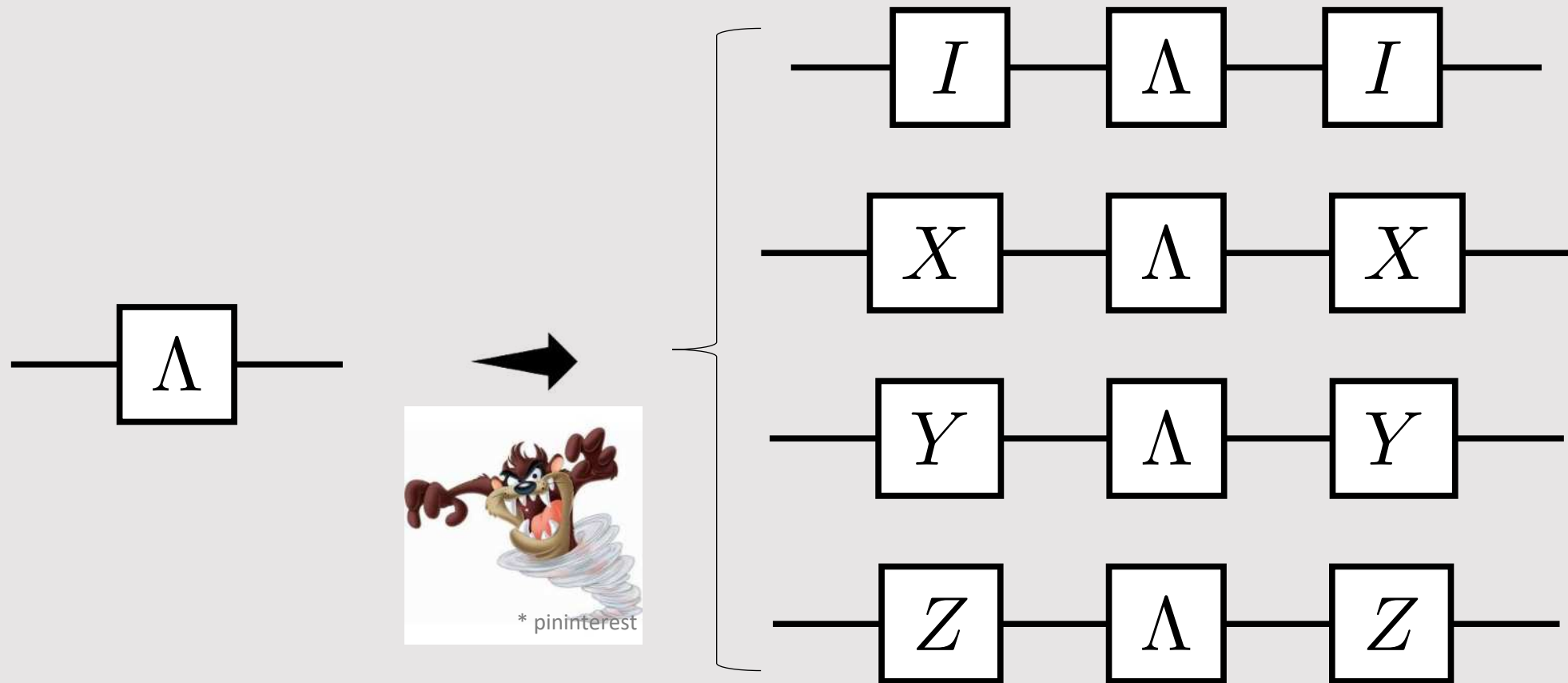
Also: <https://qiskit.org/textbook-beta/summer-school/quantum-computing-and-quantum-learning-2021>



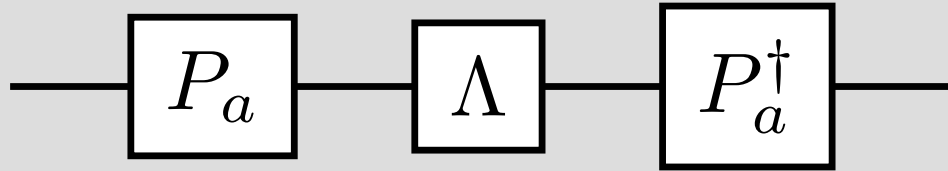
Example

Pauli twirling on a qubit

Twirl: average over instances



Twirl example on a bit flip channel



Algebraic expression of channel sequence:

$$\mathcal{P}_a \Lambda \mathcal{P}_a^\dagger = P_a^\dagger \Lambda (P_a \cdot P_a^\dagger) P_a$$

Example: Bit-flip channel

$$\begin{aligned} \Lambda(\cdot) &= (1-p) I \cdot I + p X \cdot X \\ &= (1-p) \mathcal{I} + p \mathcal{X} \end{aligned}$$

Notation

$$\begin{array}{l|l} \mathcal{I}(\cdot) = I \cdot I & \mathcal{Y}(\cdot) = Y \cdot Y \\ \mathcal{X}(\cdot) = X \cdot X & \mathcal{Z}(\cdot) = Z \cdot Z \end{array}$$

Recall $X^2 = Y^2 = Z^2 = I$

$$\begin{array}{l|l} \mathcal{I}(X) = X & \mathcal{Y}(X) = YXY = -X \\ \mathcal{X}(X) = X & \mathcal{Z}(X) = ZXZ = -X \end{array}$$

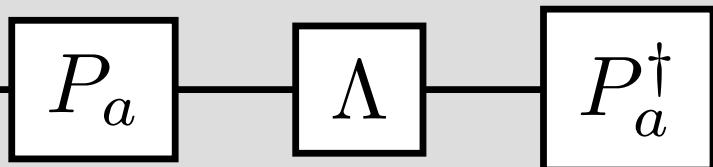
Twirl gate

$$\text{---} \boxed{I} \text{---} \quad \mathcal{I} \Lambda \mathcal{I} = \Lambda$$

$$\begin{aligned} \text{---} \boxed{X} \text{---} \quad \mathcal{X} \Lambda \mathcal{X} &= X \Lambda (X \cdot X) X \\ &= (1-p) XIX \cdot XIX + p XXX \cdot XXX \\ &= (1-p) \mathcal{X}(I) \cdot \mathcal{X}(I) + p \mathcal{X}(X) \cdot \mathcal{X}(X) \\ &= \Lambda \end{aligned}$$

$$\text{---} \boxed{Y} \text{---} \quad \mathcal{Y} \Lambda \mathcal{Y} = Y \Lambda (Y \cdot Y) Y$$

Twirl example on a bit flip channel



Example: Bit-flip channel

$$\begin{aligned}\Lambda(\cdot) &= (1-p) I \cdot I + p X \cdot X \\ &= (1-p) \mathcal{I} + p \mathcal{X}\end{aligned}$$

Notation

$$\begin{array}{l|l}\mathcal{I}(\cdot) = I \cdot I & \mathcal{Y}(\cdot) = Y \cdot Y \\ \mathcal{X}(\cdot) = X \cdot X & \mathcal{Z}(\cdot) = Z \cdot Z\end{array}$$

Recall $X^2 = Y^2 = Z^2 = I$

$$\begin{array}{l|l}\mathcal{I}(X) = X & \mathcal{Y}(X) = YXY = -X \\ \mathcal{X}(X) = X & \mathcal{Z}(X) = ZXZ = -X\end{array}$$

Twirl gate

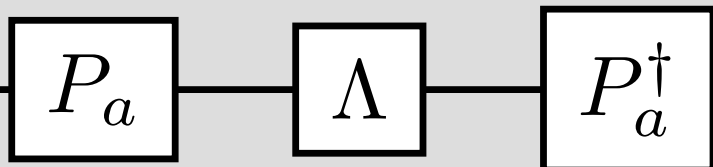
$$\text{---} \boxed{I} \text{---} \quad \mathcal{I} \Lambda \mathcal{I} = \Lambda$$

$$\begin{aligned}\text{---} \boxed{X} \text{---} \quad \mathcal{X} \Lambda \mathcal{X} &= X \Lambda (X \cdot X) X \\ &= (1-p) XIX \cdot XIX + pXXX \cdot XXX \\ &= (1-p) \mathcal{X}(I) \cdot \mathcal{X}(I) + p\mathcal{X}(X) \cdot \mathcal{X}(X) \\ &= \Lambda\end{aligned}$$

$$\begin{aligned}\text{---} \boxed{Y} \text{---} \quad \mathcal{Y} \Lambda \mathcal{Y} &= Y \Lambda (Y \cdot Y) Y \\ &= (1-p) YIY \cdot YIY + pYXY \cdot YXY \\ &= (1-p) \mathcal{Y}(I) \cdot \mathcal{Y}(I) + p\mathcal{Y}(X) \cdot \mathcal{Y}(X) \\ &= (1-p) I \cdot I + p(-X) \cdot (-X) \\ &= \Lambda\end{aligned}$$

$$\begin{aligned}\text{---} \boxed{Z} \text{---} \quad \mathcal{Z} \Lambda \mathcal{Z} &= Z \Lambda (Z \cdot Z) Z \\ &= (1-p) Z(I) \cdot Z(I) + p\mathcal{Z}(X) \cdot \mathcal{Z}(X) \\ &= \Lambda\end{aligned}$$

Twirl example on a bit flip channel



Example: Bit-flip channel

$$\begin{aligned}\Lambda(\cdot) &= (1-p) I \cdot I + pX \cdot X \\ &= (1-p) \mathcal{I} + p\mathcal{X}\end{aligned}$$

Notation

$$\begin{array}{l|l}\mathcal{I}(\cdot) = I \cdot I & \mathcal{Y}(\cdot) = Y \cdot Y \\ \mathcal{X}(\cdot) = X \cdot X & \mathcal{Z}(\cdot) = Z \cdot Z\end{array}$$

Recall $X^2 = Y^2 = Z^2 = I$

$$\begin{array}{l|l}\mathcal{I}(X) = X & \mathcal{Y}(X) = YXY = -X \\ \mathcal{X}(X) = X & \mathcal{Z}(X) = ZXZ = -X\end{array}$$

Twirl gate

$$\text{---} \boxed{I} \text{---} = \Lambda$$

$$\text{---} \boxed{X} \text{---} = \Lambda$$

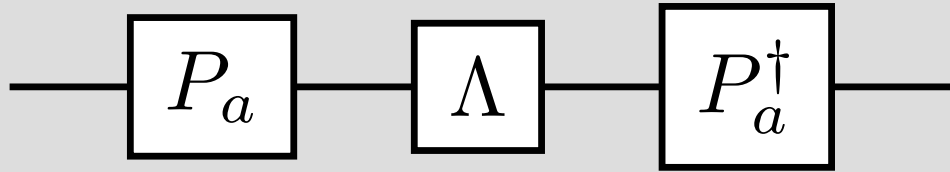
$$\text{---} \boxed{Y} \text{---} = \Lambda$$

$$\text{---} \boxed{Z} \text{---} = \Lambda$$

Average

$$\Lambda \mapsto \frac{1}{4} (\mathcal{I}\Lambda\mathcal{I} + \mathcal{X}\Lambda\mathcal{X} + \mathcal{Y}\Lambda\mathcal{Y} + \mathcal{Z}\Lambda\mathcal{Z}) = \Lambda$$

Example coherent rotation channel



Example: Coherent rotation

$$U = \exp\left(-i\frac{\theta}{2}X\right) = \cos\left(\frac{\theta}{2}\right)I - i\sin\left(\frac{\theta}{2}\right)X$$

$$\begin{aligned}\Lambda(\cdot) &= U \cdot U^\dagger \\ &= \left[\cos\left(\frac{\theta}{2}\right)\right]^2 I \cdot I + \left[\sin\left(\frac{\theta}{2}\right)\right]^2 X \cdot X \\ &\quad + \frac{i}{2}(\sin(\theta)I \cdot X - \sin(\theta)X \cdot I)\end{aligned}$$

$$\begin{aligned}&= |I\rangle\rangle\langle\langle I| + |X\rangle\rangle\langle\langle X| \\ &\quad + \cos(\theta)(|Y\rangle\rangle\langle\langle Y| + |Z\rangle\rangle\langle\langle Z|) \\ &\quad + \sin(\theta)(|Z\rangle\rangle\langle\langle Y| - |Y\rangle\rangle\langle\langle Z|)\end{aligned}$$

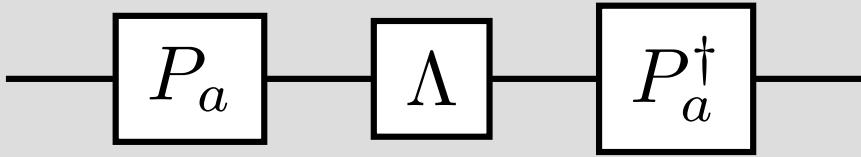
Chi matrix

	I	X	Y	Z
I	$\cos\left[\frac{\theta}{2}\right]^2$	$\frac{1}{2}i\sin[\theta]$	0	0
X	$-\frac{1}{2}i\sin[\theta]$	$\sin\left[\frac{\theta}{2}\right]^2$	0	0
Y	0	0	0	0
Z	0	0	0	0

Pauli transfer matrix

	I	X	Y	Z
I	1	0	0	0
X	0	1	0	0
Y	0	0	$\cos[\theta]$	$-\sin[\theta]$
Z	0	0	$\sin[\theta]$	$\cos[\theta]$

Example coherent rotation channel



Example: Coherent rotation

$$\Lambda(\cdot) = U \cdot U^\dagger$$

Chi matrix

	I	X	Y	Z
I	$\cos^2\left[\frac{\theta}{2}\right]$	$\frac{1}{2}i \sin[\theta]$	0	0
X	$-\frac{1}{2}i \sin[\theta]$	$\sin^2\left[\frac{\theta}{2}\right]$	0	0
Y	0	0	0	0
Z	0	0	0	0

Pauli transfer matrix

	I	X	Y	Z
I	1	0	0	0
X	0	1	0	0
Y	0	0	$\cos[\theta]$	$-\sin[\theta]$
Z	0	0	$\sin[\theta]$	$\cos[\theta]$

Twirl gate

$$\text{---} \boxed{I} \text{---} \quad \mathcal{I} \Lambda \mathcal{I} = \Lambda$$

$$\text{---} \boxed{X} \text{---} \quad \mathcal{X} \Lambda \mathcal{X} = \cos^2\left(\frac{\theta}{2}\right) \mathcal{X}(I) \cdot \mathcal{X}(I) + \sin^2\left(\frac{\theta}{2}\right) \mathcal{X}(X) \cdot \mathcal{X}(X) \\ + \frac{i}{2} (\sin(\theta) \mathcal{X}(I) \cdot \mathcal{X}(X) - \sin(\theta) \mathcal{X}(X) \cdot \mathcal{X}(I))$$

$\mathcal{X}(I) = I$
 $\mathcal{X}(X) = X$

$$= \Lambda$$

$$\text{---} \boxed{Y} \text{---} \quad \mathcal{Y} \Lambda \mathcal{Y} = \cos^2\left(\frac{\theta}{2}\right) \mathcal{Y}(I) \cdot \mathcal{Y}(I) + \sin^2\left(\frac{\theta}{2}\right) \mathcal{Y}(X) \cdot \mathcal{Y}(X) \\ + \frac{i}{2} (\sin(\theta) \mathcal{Y}(I) \cdot \mathcal{Y}(X) - \sin(\theta) \mathcal{Y}(X) \cdot \mathcal{Y}(I))$$

$\mathcal{Y}(I) = I$
 $\mathcal{Y}(X) = -X$

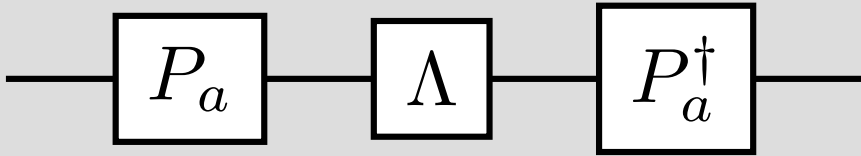
$$=$$

$$\cos^2\left(\frac{\theta}{2}\right) I \cdot I + \sin^2\left(\frac{\theta}{2}\right) X \cdot X \\ + \frac{i}{2} (-\sin(\theta) I \cdot X + \sin(\theta) X \cdot I)$$

$$\text{---} \boxed{Z} \text{---} \quad \mathcal{Z} \Lambda \mathcal{Z} = \text{same} \nearrow$$

$\mathcal{Z}(I) = I$
 $\mathcal{Z}(X) = -X$

Example coherent rotation channel



Example: Coherent rotation

$$\Lambda(\cdot) = U \cdot U^\dagger$$

Chi matrix

	I	X	Y	Z
I	$\text{Cos}[\frac{\theta}{2}]^2$	$\frac{1}{2} i \text{Sin}[\theta]$	0	0
X	$-\frac{1}{2} i \text{Sin}[\theta]$	$\text{Sin}[\frac{\theta}{2}]^2$	0	0
Y	0	0	0	0
Z	0	0	0	0

Pauli transfer matrix

	I	X	Y	Z
I	1	0	0	0
X	0	1	0	0
Y	0	0	$\text{Cos}[\theta]$	$-\text{Sin}[\theta]$
Z	0	0	$\text{Sin}[\theta]$	$\text{Cos}[\theta]$

Twirl



Twirl average

$$\Lambda \mapsto \frac{1}{4} (\mathcal{I}\Lambda\mathcal{I} + \mathcal{X}\Lambda\mathcal{X} + \mathcal{Y}\Lambda\mathcal{Y} + \mathcal{Z}\Lambda\mathcal{Z})$$

$\chi =$

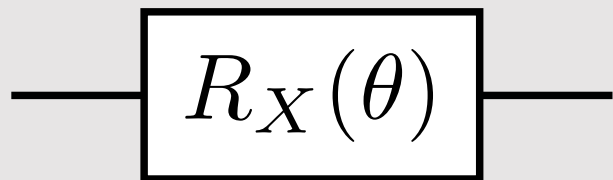
	I	X	Y	Z
I	$\text{Cos}[\frac{\theta}{2}]^2$	0	0	0
X	0	$\text{Sin}[\frac{\theta}{2}]^2$	0	0
Y	0	0	0	0
Z	0	0	0	0

Diagonal

PTM =

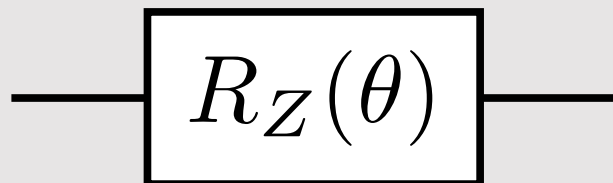
	I	X	Y	Z
I	1	0	0	0
X	0	1	0	0
Y	0	0	$\text{Cos}[\theta]$	0
Z	0	0	0	$\text{Cos}[\theta]$

Example PTM for coherent noise



$$\text{PTM}[R_X(\theta)] = \begin{matrix} & \begin{matrix} I & X & Y & Z \end{matrix} \\ \begin{matrix} I \\ X \\ Y \\ Z \end{matrix} & \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \cos \theta & -\sin \theta \\ 0 & 0 & \sin \theta & \cos \theta \end{pmatrix} \end{matrix}$$

Note, for other gates, permute indices
Same story



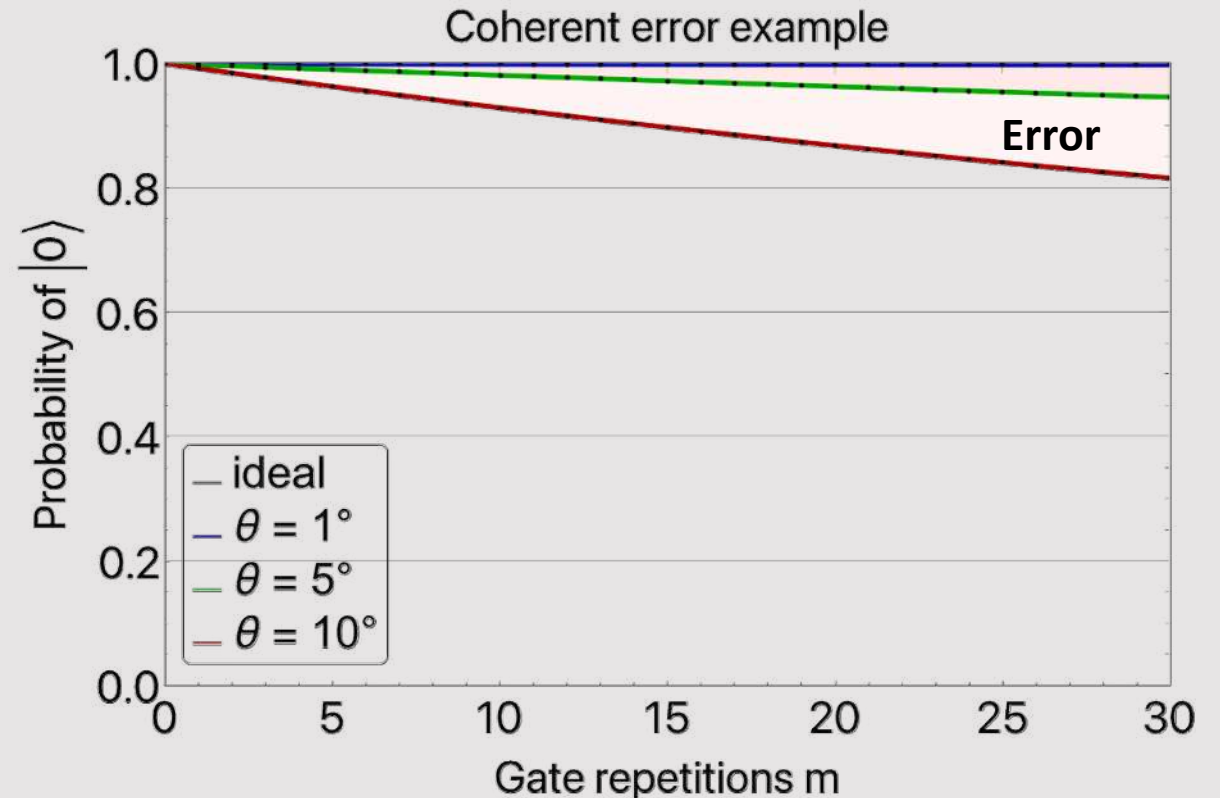
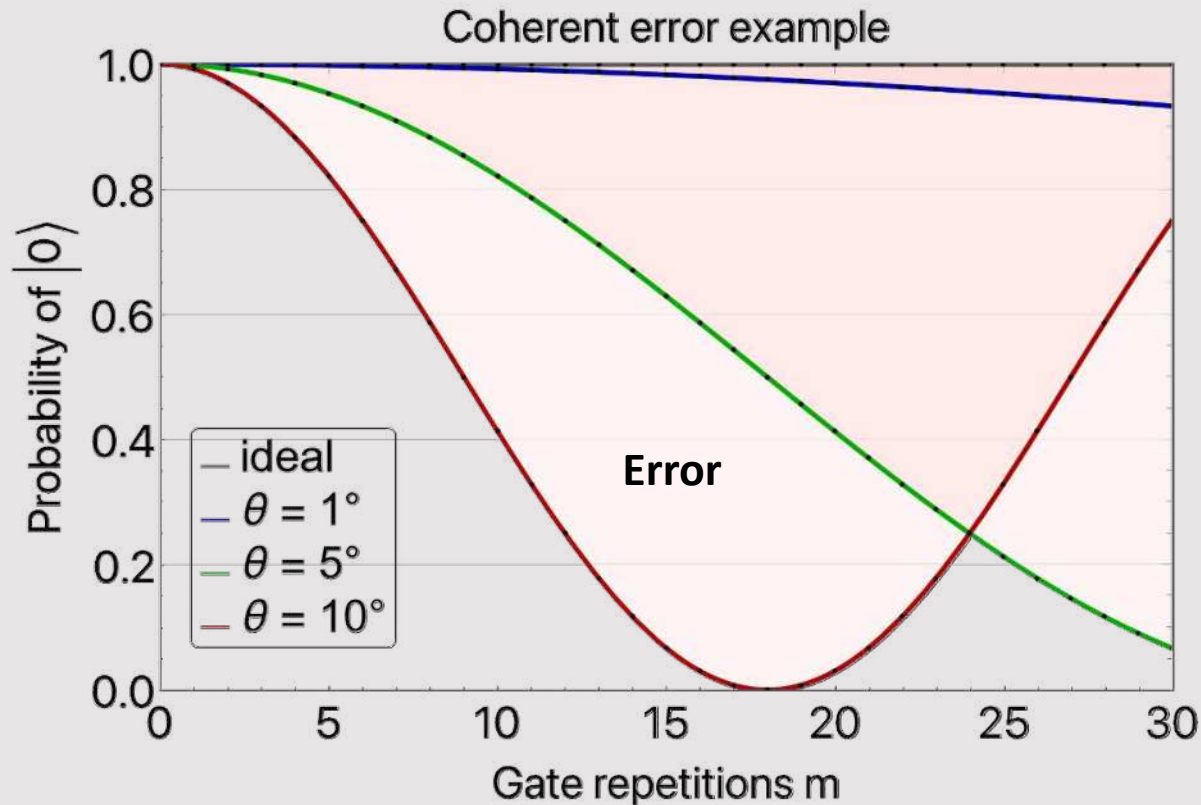
$$\text{PTM}[R_Z(\theta)] = \begin{matrix} & \begin{matrix} I & X & Y & Z \end{matrix} \\ \begin{matrix} I \\ X \\ Y \\ Z \end{matrix} & \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta & 0 \\ 0 & \sin \theta & \cos \theta & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \end{matrix}$$

Example: Coherent over rotation

Suppose we meant to do an identity gate, but instead had a small X over rotation of angle theta

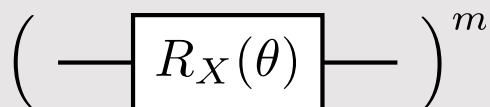
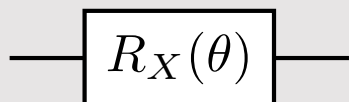
$$\left(\text{---} \boxed{R_X(\theta)} \text{---} \right)^m$$

$$\left(\text{---} \boxed{P_{ai}} \boxed{R_X(\theta)} \boxed{P_{ai}^c} \text{---} \right)^m$$



Example: Coherent over rotation

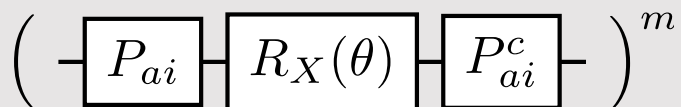
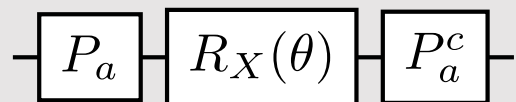
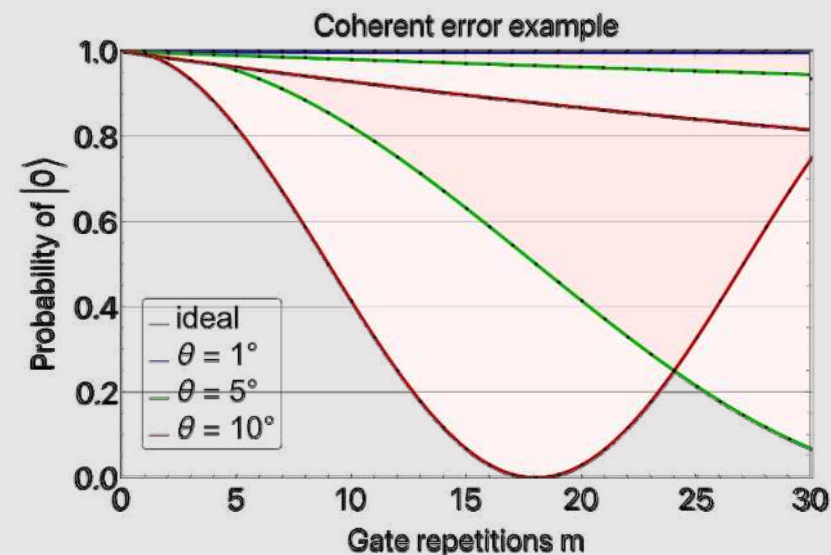
Suppose we meant to do an identity gate, but instead had a small X over rotation of angle theta



PTM

$$(\hat{R}) = \begin{matrix} & I & X & Y & Z \\ \begin{matrix} I \\ X \\ Y \\ Z \end{matrix} & \begin{pmatrix} 1 & & & \\ & 1 & & \\ & & \cos(\theta) & -\sin(\theta) \\ & & \sin(\theta) & \cos(\theta) \end{pmatrix} \end{matrix}$$

$$(\hat{R}^m) = \begin{matrix} & I & X & Y & Z \\ \begin{matrix} I \\ X \\ Y \\ Z \end{matrix} & \begin{pmatrix} 1 & & & \\ & 1 & & \\ & & \cos(m\theta) & -\sin(m\theta) \\ & & \sin(m\theta) & \cos(m\theta) \end{pmatrix} \end{matrix}$$



$$(\mathcal{T}\hat{R}) = \begin{matrix} & I & X & Y & Z \\ \begin{matrix} I \\ X \\ Y \\ Z \end{matrix} & \begin{pmatrix} 1 & & & \\ & 1 & & \\ & & \cos(\theta) & \\ & & & \cos(\theta) \end{pmatrix} \end{matrix}$$

$$([\mathcal{T}\hat{R}]^m) = \begin{matrix} & I & X & Y & Z \\ \begin{matrix} I \\ X \\ Y \\ Z \end{matrix} & \begin{pmatrix} 1 & & & \\ & 1 & & \\ & & [\cos(\theta)]^m & \\ & & & [\cos(\theta)]^m \end{pmatrix} \end{matrix}$$

$$\langle Z \rangle_{\text{noisy}} - \langle Z \rangle_{\text{ideal}} =$$

Coherent error - quadratic

$$\cos(n\theta) - 1 \approx -\frac{n^2\theta^2}{2} + \mathcal{O}(\theta^4)$$

Twirl error - linear

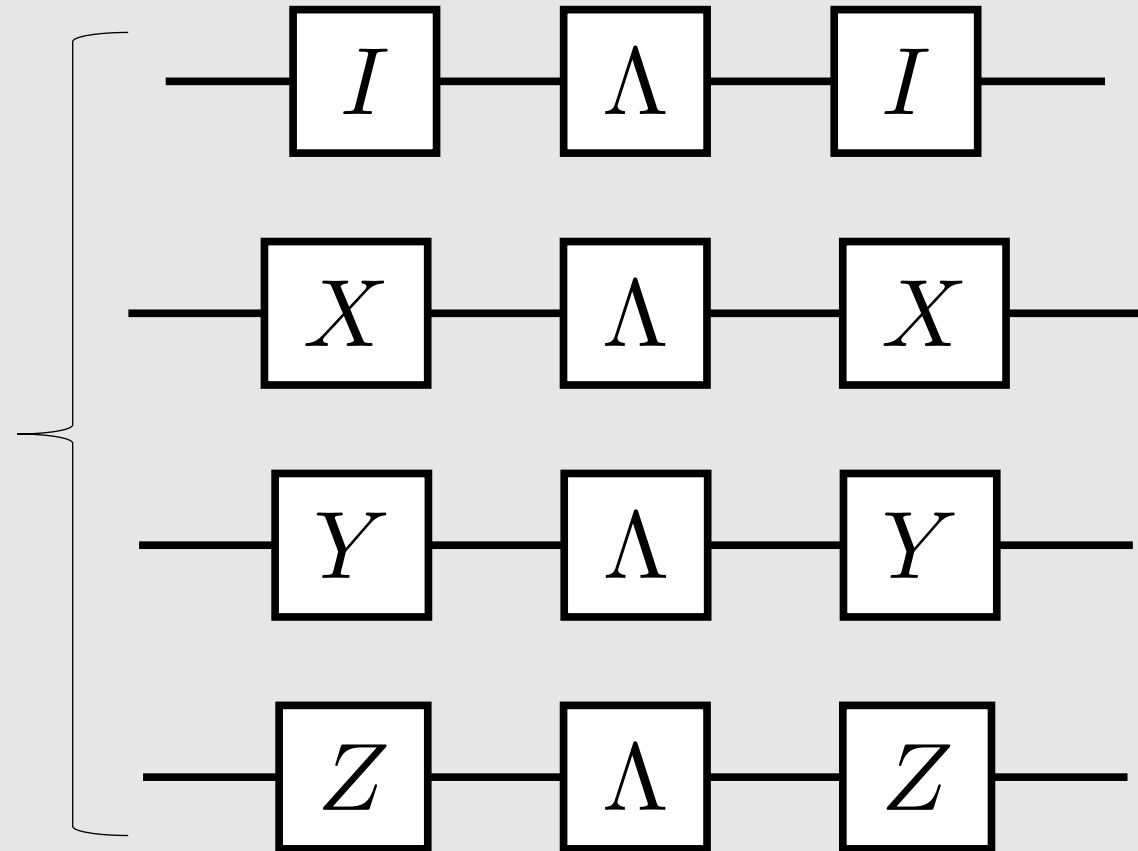
$$[\cos(\theta)]^m - 1 \approx -\frac{n\theta^2}{2} + \mathcal{O}(\theta^4)$$

Twirl general single qubit channel

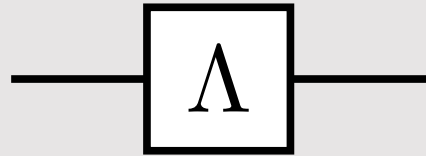


In terms of PTM

	I	X	Y	Z
I	f_{II}	f_{IX}	f_{IY}	f_{IZ}
X	f_{XI}	f_{XX}	f_{XY}	f_{XZ}
Y	f_{YI}	f_{YX}	f_{YY}	f_{YZ}
Z	f_{ZI}	f_{ZX}	f_{ZY}	f_{ZZ}



Twirl general single qubit channel



In terms of PTM

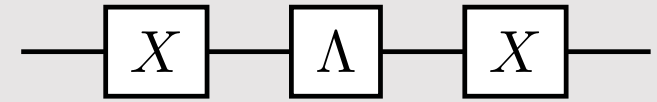
	I	X	Y	Z
I	f_{II}	f_{IX}	f_{IY}	f_{IZ}
X	f_{XI}	f_{XX}	f_{XY}	f_{XZ}
Y	f_{YI}	f_{YX}	f_{YY}	f_{YZ}
Z	f_{ZI}	f_{ZX}	f_{ZY}	f_{ZZ}



	I	X	Y	Z
I	f_{II}	f_{IX}	f_{IY}	f_{IZ}
X	f_{XI}	f_{XX}	f_{XY}	f_{XZ}
Y	f_{YI}	f_{YX}	f_{YY}	f_{YZ}
Z	f_{ZI}	f_{ZX}	f_{ZY}	f_{ZZ}



	I	X	Y	Z
I	f_{II}	$-f_{IX}$	f_{IY}	$-f_{IZ}$
X	$-f_{XI}$	f_{XX}	$-f_{XY}$	f_{XZ}
Y	f_{YI}	$-f_{YX}$	f_{YY}	$-f_{YZ}$
Z	$-f_{ZI}$	f_{ZX}	$-f_{ZY}$	f_{ZZ}

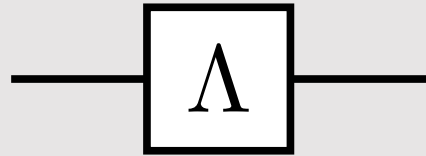


	I	X	Y	Z
I	f_{II}	f_{IX}	$-f_{IY}$	$-f_{IZ}$
X	f_{XI}	f_{XX}	$-f_{XY}$	$-f_{XZ}$
Y	$-f_{YI}$	$-f_{YX}$	f_{YY}	f_{YZ}
Z	$-f_{ZI}$	$-f_{ZX}$	f_{ZY}	f_{ZZ}



	I	X	Y	Z
I	f_{II}	$-f_{IX}$	$-f_{IY}$	f_{IZ}
X	$-f_{XI}$	f_{XX}	f_{XY}	$-f_{XZ}$
Y	$-f_{YI}$	f_{YX}	f_{YY}	$-f_{YZ}$
Z	f_{ZI}	$-f_{ZX}$	$-f_{ZY}$	f_{ZZ}

Twirl general single qubit channel



In terms of PTM

	I	X	Y	Z
I	f_{II}	f_{IX}	f_{IY}	f_{IZ}
X	f_{XI}	f_{XX}	f_{XY}	f_{XZ}
Y	f_{YI}	f_{YX}	f_{YY}	f_{YZ}
Z	f_{ZI}	f_{ZX}	f_{ZY}	f_{ZZ}

Average

	I	X	Y	Z
I	f_{II}	$\circ$	$\circ$	$\circ$
X	$\circ$	f_{XX}	$\circ$	$\circ$
Y	$\circ$	$\circ$	f_{YY}	$\circ$
Z	$\circ$	$\circ$	$\circ$	f_{ZZ}



* pikisuperstar

Refresher

More general

Pauli gates & mixed states